ECDLP formal work against the CFSG failure modes

2026-10-29

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Source: KN-LIT-63f33d, Axiom Math's agent-built Lean proof of the classification of finite simple groups. It reports four failure modes:

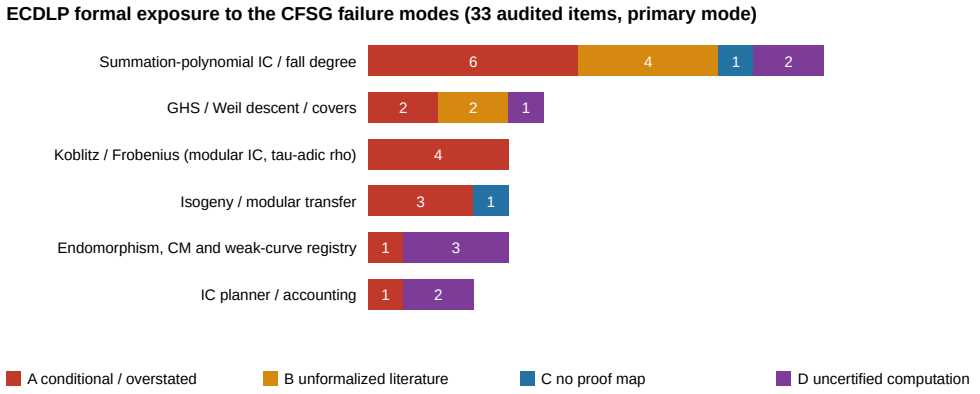
- A. A theorem is cited for more than its hypotheses give.
- B. A published lemma is taken as given without checking.
- C. A multi-lemma effort has no dependency map.
- D. A finite computation has no independently checkable certificate.

Scope. We ran a read-only audit of ECDLP formal work in three repositories:

- crypto-autoresearcher: formal/, frontier rows and leads;
- crypto: 34 Lake projects across the crypto and cryptanalysis research trees, counted together;
- cryptanalysis.

Two independent audit sessions did the work. The coordinating session spot-checked five items against the source: frontier rows KR-IC-f5c584 and KR-RHO-755e35, CubicGluing, S3Roots and CyclicWindow. All five hold.

What this report is. An exposure map, not a review decision. It changes no hypothesis, frontier row or lead. Every proposed fix below needs the Coordinator, or the owner of the cited record.



Exposure by ECDLP area

Headline

- Lean hygiene is clean. No sorry, admit or custom axiom appears in any audited project. The only native_decide is crypto/research/ic_tree_split_structured_v_20261011/lean/TreeSplit/Replay.lean:249, and it is disclosed. In nearly every case the Lean file headers state their own scope correctly.
- The exposure sits one layer up, in frontier rows, leads, README headings and "proved" cells. These state results more broadly than the Lean or the sources support. This is mode A, the failure an axiom audit cannot see.
- Many literature results are filed as proven from an abstract read only. They include Semaev's theorem, Diem's complexity results, the GHS and Iijima–Momose–Chao results, and JMV. This is mode B.

Areas affected, ranked by exposure

1. Summation-polynomial index calculus and first-fall degree (13 items).
   - KR-IC-f5c584 claims first fall degree 2 for S_3, has status proven, and forecloses "first fall degree 3 on S3 descent". Its sources are abstract or secondary reads. The finite control in formal/targets/corrections/2026-10-25-semaev-s3-degree-two.md finds no new class at degree 2 and one at degree 3. The row does not cite that correction.
   - LEAD-ICE-first-fall-degree-assumptions still asks to discharge the bridge that was already refuted.
   - Semaev's correspondence, that S_m vanishes exactly when ±P sum to O, is not formalized anywhere. crypto/papers/rootfinding-decomposition/lean/S3Roots.lean defines its own chord formulas and never proves them equal to Mathlib's group law.
   - KN-FIND-a8990a claims (Z/2)^{m-2} monodromy in every characteristic. Its corroboration appears to cover F_p only.
   - The Nagao Lemma 4 erratum (KN-FIND-936151) is a case where this program caught a published error. Its counterexample exists only as a Python certificate.
   - The 282 d_F values in crypto/.../fall_degree_bounds_20261008 have no rank certificates.
   - None of the 89 formal targets uses proof_map. The S_3 lane was split into seven targets after a hidden prerequisite surfaced, the map from equations to row space.
2. GHS / Weil descent / cover attacks (5 items).

- The proven foreclosures KR-IC-64659b, KR-IC-5ea2f8 and KR-IC-0d2021 rest on abstract reads only. They bear on LEAD-TRF-ghs-binary-composite and LEAD-ICB-composite-degree-c2pnb.
- ghs_cubic_hyperelliptic_genus3 marks “genus 3 by subcover sum” as proved. Lean checks only the bitmask arithmetic; Kani–Rosen is assumed.
- In the weak-curve database, ghsGenusForType is a definition, not a theorem.
- cryptanalysis/experiments/ghs-sweeper/formal/CubicGluing.lean:49 proves $3 \cdot 4 = 3^3$ in F_5. The comment says this is det M = q^3, which is false over $\mathbb{Z}$.
- The Enge cover Lean project was never built.

3. Koblitz / Frobenius (4 items). These cover modular index calculus and τ-adic rho.
   - koblitz_modular_ic states “no degree-131 cyclic carrier below genus 394” unconditionally in its heading. CyclicWindow.lean takes the spectral bound hSpectral as a hypothesis.
   - The Q34 exclusion depends on a descent bridge that is only written out.
   - The √m gain in KR-RHO-482ed3 assumes that τ(G) = [λ]G.
   - LEAD-RHO-automorphism-class-walks says “strong rho correct (Lean)”. The Lean models the group as Z_r; the curve and arithmetic bridge is still open.
4. Isogeny / modular transfer (4 items).
   - KR-RHO-755e35 (JMV) is filed as proven without naming GRH.
   - Only 19 of the 84 reviewed declarations received a fidelity review. The transport theorems assume reverse composition and coprimality.
   - Hecke-432 bolds “trivial torsion”. The fact that torsion injects into a good reduction enters only as a hypothesis.
   - About 70 Hecke modules have no proof map across them.
5. Endomorphism, CM and the weak-curve registry (4 items).
   - The “fundamental D_K” claims in DkExact* rest on PARI isprime, with no archived primality certificates.
   - The Hasse bound and point orders are premises (13,219 orders, checked in Python only).
   - The witness search in CubicCM53 is bounded.
6. IC planner and accounting (3 items).
   - ic_planner_exhaustive is titled “proven in Lean”, and a leaderboard row cites it. Its key premise, that the Rust collector uses only four group operations, was checked by reading the code.

Proposed actions, cheapest first

These are proposals for the Coordinator. None has been applied.

Action	Closes	Cost
Downgrade KR-IC-f5c584 from proven and cite the 2026-10-25 correction; update LEAD-ICE-first-fall-degree-assumptions	Area 1, row-level A	record edit
Name GRH in KR-RHO-755e35; mark KR-IC-5931ee heuristic; state the conditions in KR-IC-fbdb61	Area 4/1, mode A	record edit
Requalify README headings: “conditional” (CyclicWindow), “formula arithmetic, Kani–Rosen assumed” (GHS genus 3), “det ≡ q^3 mod 5” (CubicGluing), “walk algebra over Z_r” (rho lead), “arithmetic of reported sums” (IC planner title)	Areas 2, 3, 6	doc edit
Add proof_map to the semaev-s3-*, lastfall and first-fall targets, chained through finite-equation semantics	Area 1, mode C	spec edit
Retroactive hypothesis-delta pass over targets cited by findings, Semaev and Endosweep first	Areas 1, 5	review batch
Encode the Nagao Lemma 4 witnesses as a Lean decide theorem; Lean decide or a Sage check that S_3 vanishes on all triples of one small curve over F_p and F_{2^k}	Area 1, modes B/D	small Lean
Exhaustive S_3/S_4 monodromy check over F_{2^k} and F_{3^k}	KN-FIND-a8990a, mode B	small computation
Full-text reads recording the exact hypotheses of Semaev, Diem, GHS/IMC and JMV, with verification_state upgraded	Areas 1, 2, 4, mode B	reading
Archive ECPP/Pratt primality certificates for DkExact and the Oakley primes; give the formal gate a per-declaration #print axioms log	Area 5, mode D	tooling

Method and limits

- Who did the work. Two read-only agent sessions, one for crypto-autoresearcher and one for crypto and cryptanalysis. Each was given the four failure modes and the project list.
- How each item is backed. Every table row in their reports says whether the auditor read the Lean statement or inferred the gap from documentation. Three items are inferred, not read: the F_p-only scope of KN-FIND-6edecd, the transport and rho Lean, and the Boolean-parity certificates.
- What was not read in depth. projective_law_followup (494 theorems).
- Missing atlas summary. tools/lead_atlas.py --brief timed out, so the lead atlas’s change summary was not consulted.

- No experiment, no performance claim. Nothing was run, and no performance number is claimed, so no benchmarks/ entry applies.